

PRANJAL NARWAL

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Rohtak , Haryana, India

SUMMARY

Results-driven Full-Stack Software Engineer with a strong foundation in building and shipping AI-powered, real-time, production-grade web applications. Proficient in the complete development lifecycle — from system design and API architecture to cloud deployment and performance optimisation. Demonstrated ability to independently engineer complex systems using React, TypeScript, Python, Node.js, PostgreSQL, and Redis. Completed over 200 algorithmic challenges on LeetCode demonstrating consistent problem-solving ability and strong computational thinking.

EDUCATION

Manipal University Jaipur Jaipur, Rajasthan, India
Bachelor of Technology — Computer Science and Engineering Aug 2023 – May 2027 (Expected)
– Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

EXPERIENCE

LaunchEd Global EdTech Bengaluru, Karnataka (Remote)
Brand Executive — Campus Ambassador Dec 2025 – May 2026
– Designed and executed targeted digital marketing campaigns to drive student registrations for the company's internship programme, applying data-driven outreach strategies across social media platforms.
– Managed end-to-end campaign operations on campus, coordinating communication between the organisation and the student community to meet performance milestones.

PROJECTS

Dora Travel Agency — *Python, Flask, JavaScript, Three.js, Leaflet.js, REST APIs, Vercel* | [Live Demo](#) | [GitHub](#)
– Engineered a production-ready, full-stack AI travel intelligence platform covering **40+ countries**, capable of generating structured multi-day itineraries, cultural advisories, emergency contacts, and destination briefs from a single user query.
– Built an interactive **3D globe** using Three.js and Globe.gl featuring animated flight arcs, zoom-to-destination transitions, and real-time destination pin rendering, significantly improving user engagement and visual storytelling.
– Architected a Flask REST API integrating **5 external data sources** — Open-Meteo (weather + geocoding + AQI), Overpass/OpenStreetMap, and Frankfurter (live FX rates) — across **11 real-time intelligence widgets** including currency converter, SIM card guide, scam advisories, and daily budget estimator.
– Delivered a **multilingual interface in 4 languages** (English, Hindi, Spanish, French) and deployed serverlessly to Vercel, achieving zero-downtime production availability with automatic scaling.

Dora Sheet — *TypeScript, React 18, NestJS, Socket.IO, PostgreSQL, Redis, Prisma, Vercel* | [Live Demo](#) | [GitHub](#)
– Designed and shipped a **real-time collaborative spreadsheet platform** — comparable to Google Sheets — featuring live multi-user presence, cursor tracking, invitation-based rooms, and host-controlled join approvals, all powered by Socket.IO WebSocket infrastructure.
– Implemented an **AI assistant with multi-tool function calling** (`apply_formula`, `fill_data`, `format_cells`, `sort_data`, `modify_structure`) that directly manipulates spreadsheet cells from natural language commands, with an Approve & Apply confirmation layer for user safety.
– Built a **full spreadsheet formula engine** from scratch, supporting SUM, AVERAGE, IF, and VLOOKUP; added Excel (.xlsx) import via SheetJS, undo/redo history, version snapshots, and find-and-replace — covering core spreadsheet functionality end-to-end.
– Architected a **production monorepo** with NestJS (API) and React 18/Vite (client), using Prisma ORM with PostgreSQL for persistent data and Redis for distributed cell-locking; deployed across Vercel and Render with a GitHub Actions CI/CD pipeline.

ShadowCart — *TypeScript, React 19, Vite, CRXJS, IndexedDB, Brevo API, Vercel Serverless* | [GitHub](#)
– Conceived and built a **Chrome browser extension** that intercepts Add-to-Cart events across any e-commerce website (Amazon, Shopify, luxury retailers), enabling users to maintain a private unified wishlist without creating store accounts — a universal shopping layer across the web.

- Implemented a **Cooling Period reminder system** with user-configurable delay timers and automated transactional email notifications via the Brevo API, reducing impulsive purchasing behaviour through behavioural design.
- Adopted a **local-first, privacy-preserving data architecture** using IndexedDB for all shopping history, ensuring zero third-party data transmission and complete user ownership of data.
- Packaged and distributed the extension using Vite with CRXJS for Chrome deployment, with a **Vercel Serverless Functions** backend handling secure email trigger logic and a spending pattern tracker to surface user behaviour insights.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, TypeScript, JavaScript (ES6+), HTML5, CSS3, SQL

Frontend: React 18/19, Vite, Tailwind CSS, Three.js, Leaflet.js, Globe.gl, Zustand, CRXJS

Backend: Node.js, NestJS, Flask, REST API Design, Socket.IO, WebSockets, Serverless Functions

Databases & Storage: PostgreSQL, Redis, IndexedDB, Prisma ORM

Cloud & DevOps: Vercel, Render, Git, GitHub, GitHub Actions (CI/CD), Postman, VS Code

Core Concepts: Data Structures & Algorithms, System Design, Real-Time Systems, AI Function Calling, Browser Extension Development, Multilingual UI, Cloud Deployment

ACHIEVEMENTS & RECOGNITION

- **International Innovation Challenge 2.0** — Competed in a national level innovation competition, cleared Round 1 and advanced to Round 2 out of a large pool of participants across India.
- **Completed 200+ Algorithmic Challenges on LeetCode** — Consistently practised Data Structures & Algorithms covering Arrays, Trees, Graphs, Dynamic Programming, and Sliding Window techniques, demonstrating strong problem-solving and computational thinking.
- **3 Full-Stack Applications Shipped to Production** — Independently architected, built, and deployed three complex production-grade applications as a third-year undergraduate student, each integrating AI, real-time collaboration, or browser-level engineering.